

# Bounded validation report

System: synthetic\_three\_element\_custom\_manifest

Status: passed

Validated scope: Software input-substitution test using deterministic synthetic tables; no material claim.

software\_configurability: demonstrated\_for\_bundled\_inputs

within\_manifold\_interpolation: validated\_internal

binary\_energy\_rank\_consistency: observed\_reference\_set

new\_host\_or\_prototype\_transferability: not\_evaluated

nonmetal\_compound\_transferability: not\_evaluated

synthesizability: not\_predicted

This report does not predict synthesizability.

# Fixed-manifold validation

Module: renamed\_composition\_module

Composition rows: 15

Mixing elements: A, B, C

Pair parameters: 3

Training R2: 1.000000000

Training RMSE: 0.000000 arbitrary energy unit

Endmember-only RMSE: 0.125779

Non-endmember LOOCV RMSE: 0.000000

Group-holdout RMSE range: 0.000000-0.000000

Interpretation: internal interpolation and ablation only.

# Energy-backend reference comparison

Module: renamed\_backend\_module

Backends: Reference backend, Candidate backend

Unit: arbitrary energy unit

all\_materials reference\_backend\_\_candidate\_backend: N=5, rho=1.0000, RMSE=0.167, top-2 overlap=2, reversals=0/10

Limitation: Deterministic synthetic data; no scientific material inference.